FRANKFURT UNIVERSITY OF APPLIED SCIENCES

Prüfprotokoll Zugversuch

Projekt Nr.: 16.06.21

Prüfnorm: DIN EN ISO 6892-1

Probe: Zugprobe DIN 50125 - B8x40

Werkstoff: S235 JR+C, S235 JR+N, C45E+C, C45E+N

Soll-Dehnungsgeschwindigkeit für R_{eH}, R_p $\dot{e}_{L_e}$ = 0,00025/s = 0,025 %/s

Soll-Dehnungsgeschwindigkeit für R_m, A, Z $\dot{e}_{L_c}$ = 0,0067/s

Messmittel

	Typ / Bezeichnung / Messmittelnummer	Messbereich / Auflösung
Prüfmaschine	Hegewald & Peschke	/
Kraftmesssystem	Kraftmessdose HBM Z4A	0 - 100 kN
Feinwegmessung	Extensom. MFA2	0 - 2 mm
Längenmessung	analog Messschieber	0 - 150 mm
Prüftemp.: RT		

III Auswertung

Probe Nr.	d_o in mm	L_o in mm	d_u in mm	L_u in mm	F_{eH} in ___	$F_{p0,2}$ in kN	F_m in ___	F_u in kN
S235 JR +C	8,00	40	4,72	48,4	/	24,25	28,30	17,75
	S_o in mm²	S_u in mm²	A in %	Z in %	R_{eH} in ___	R_{p02} in MPa	R_m in MPa	R_u in MPa
	50,2655	17,4974	21,0 (21)	65,2	/	482	563	1104

III Auswertung

Probe Nr.	d_o in mm	L_o in mm	d_u in mm	L_u in mm	F_{eH} in kN	$F_{p0,2}$ in kN	F_m in kN	F_u in kN
S235 JR +N	8,02	40	4,52	54,7	15,50	/	21,75	14,25
	S_o in mm²	S_u in mm²	A in %	Z in %	R_{eH} in MPa	R_{p02} in ___	R_m in MPa	R_u in MPa
	50,5171	16,0460	36,8	68,2	307	/	431	888

III Auswertung

Probe Nr.	d_o in mm	L_o in mm	d_u in mm	L_u in mm	F_{eH} in kN	$F_{p0,2}$ in kN	F_m in kN	F_u in kN
C45E +N	8,08	40	5,80	45,4	18,5	/	29,8	25,25
	S_0 in mm^2	S_u in mm^2	A in %	Z in %	R_{eH} in MPa	R_{p02} in___	R_m in MPa	R_u in MPa
	51,28	26,42	13,5	48,5	361	/	581	955

III Auswertung

Probe Nr.	d_o in mm	L_o in mm	d_u in mm	L_u in mm	F_{eH} in___	$F_{p0,2}$ in kN	F_m in kN	F_u in kN
C45E +C	8,04	40	6,38	43,7	/	36,4	42,0	34,25
	S_0 in mm^2	S_u in mm^2	A in %	Z in %	R_{eH} in___	R_{p02} in MPa	R_m in MPa	R_u in MPa
	50,7694		9,3	37,0	/	717	827	1071

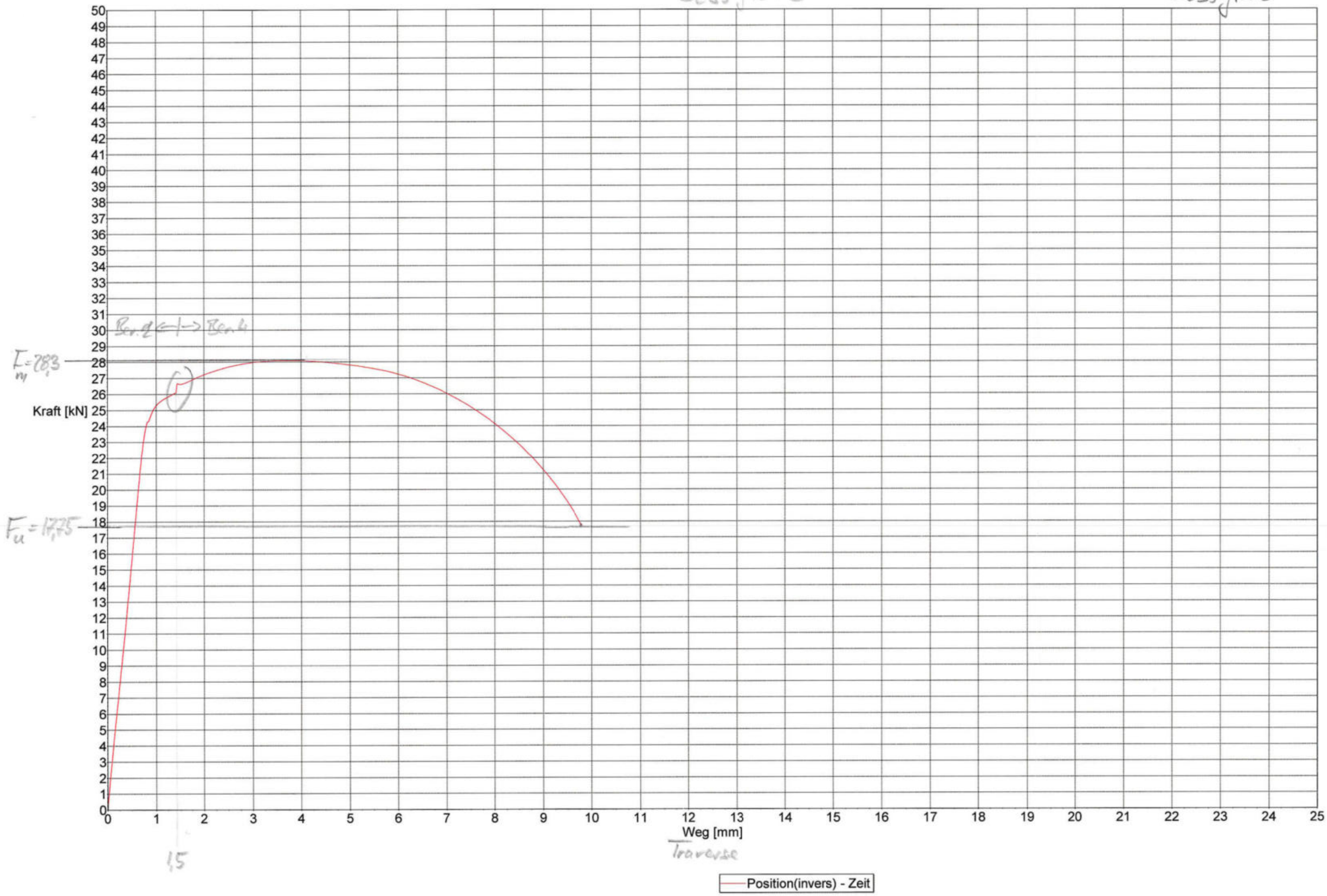
III Auswertung

Probe Nr.	d_o in___	L_o in___	d_u in___	L_u in___	F_{eH} in___	$F_{p0,2}$ in___	F_m in___	F_u in___
			A__ in___	Z in___	R_{eH} in___	R_{p02} in___	R_m in___	R_u in___

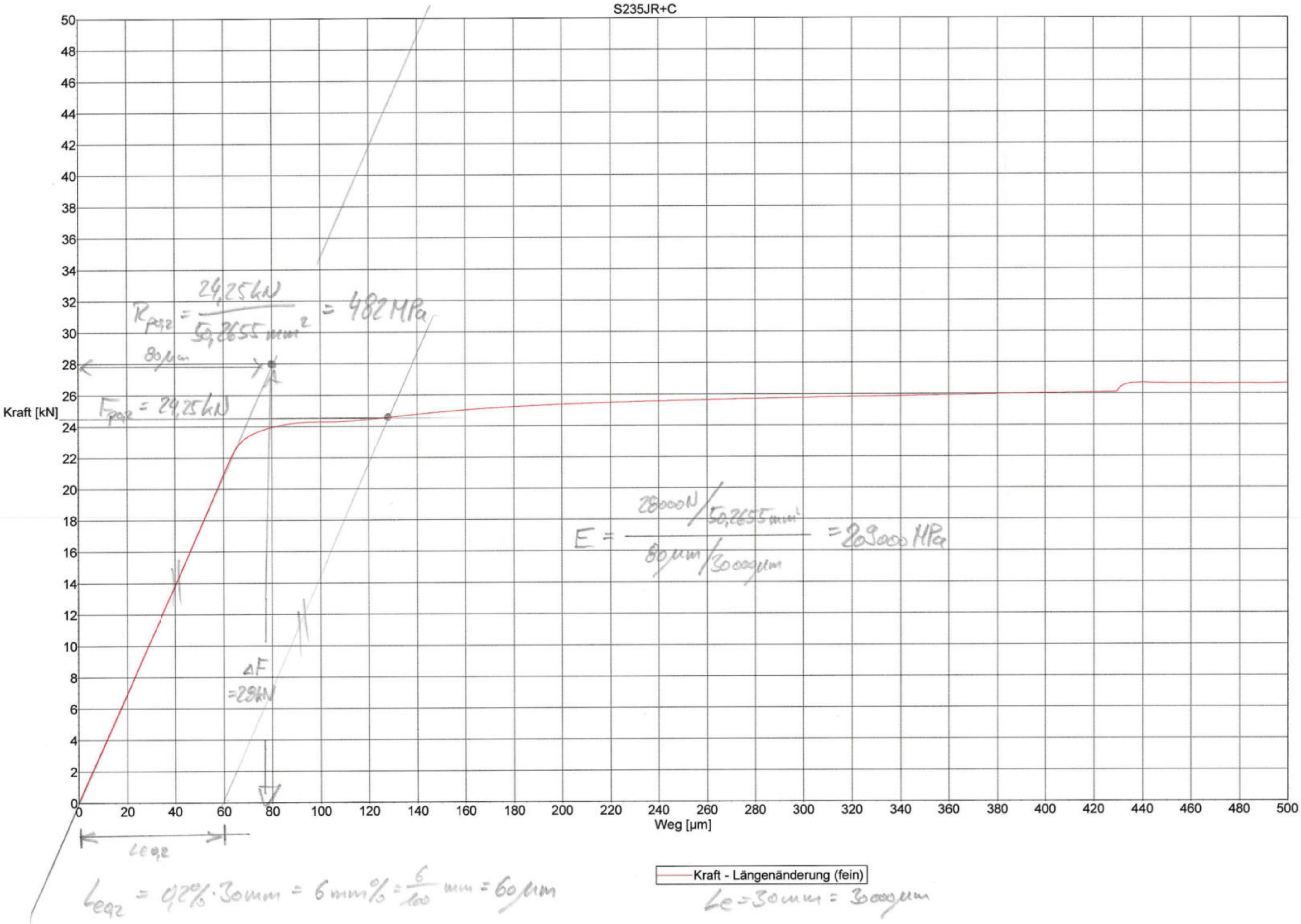
III Auswertung

Probe Nr.	d_o in___	L_o in___	d_u in___	L_u in___	F_{eH} in___	$F_{p0,2}$ in___	F_m in___	F_u in___
			A__ in___	Z in___	R_{eH} in___	R_{p02} in___	R_m in___	R_u in___

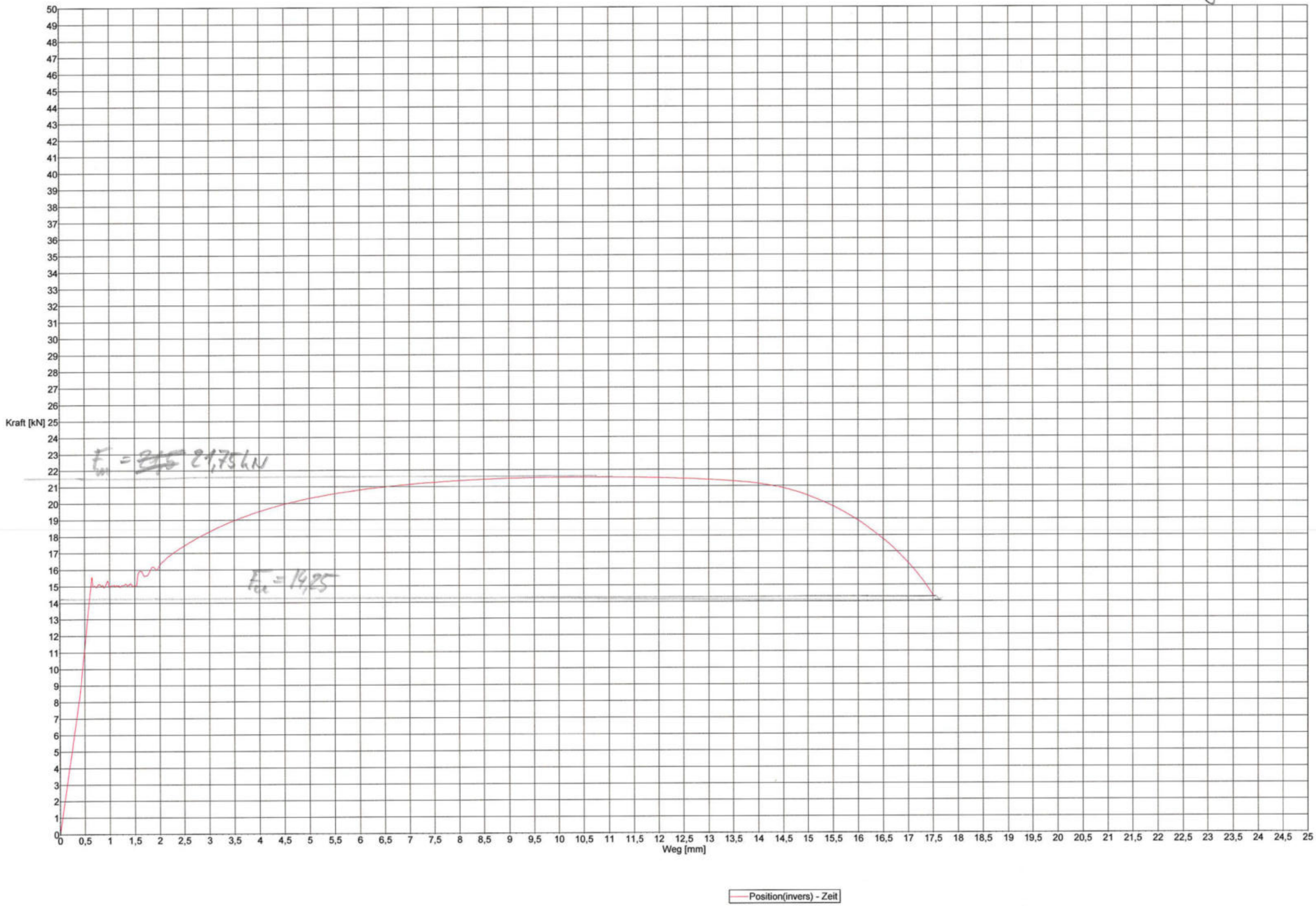
S235 JR+C

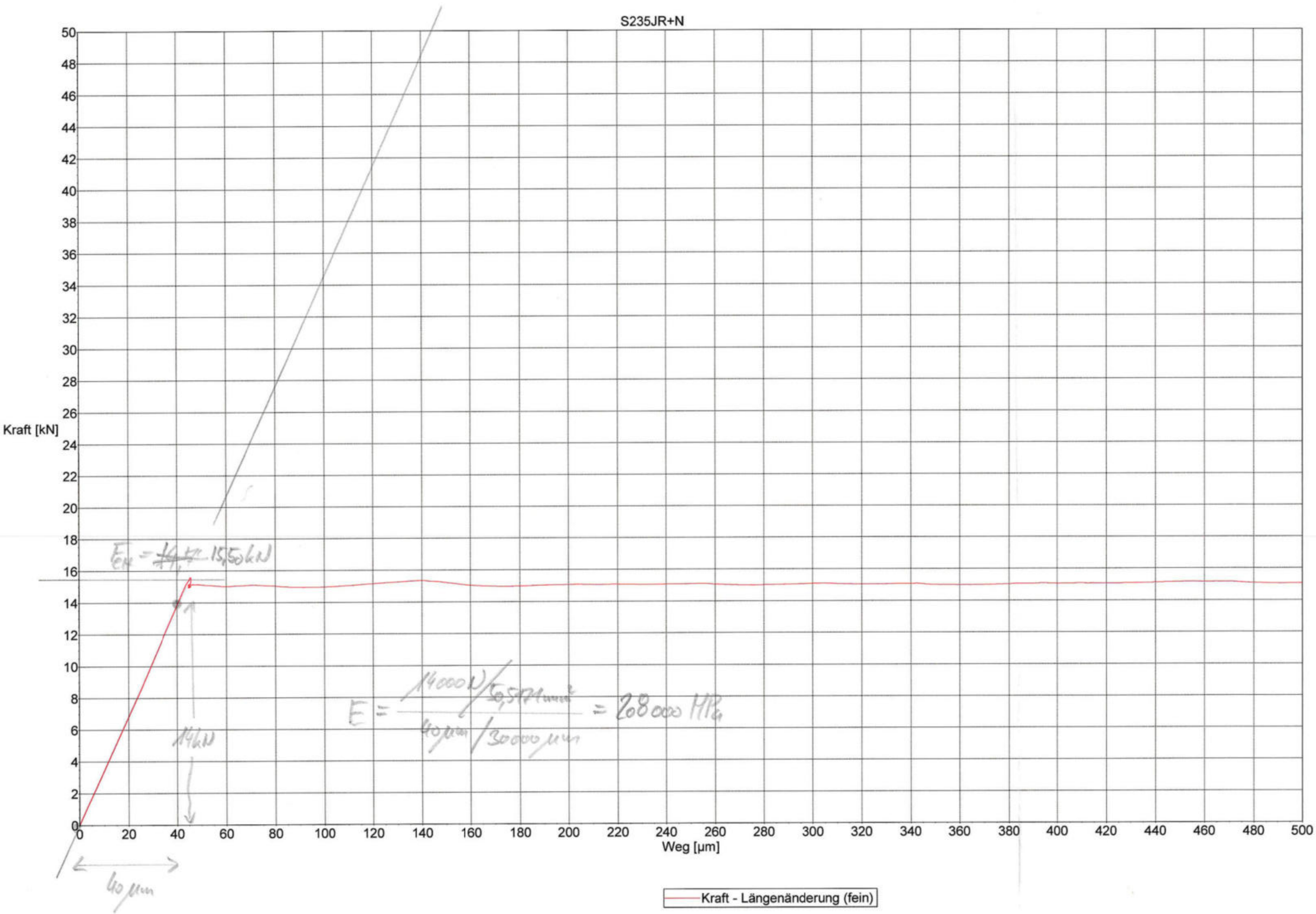


S235 JR+C
Kraft [kN]
Weg [mm]
Traverse
F_m = 28,3
F_u = 17,75
1,5
Position(invers) - Zeit

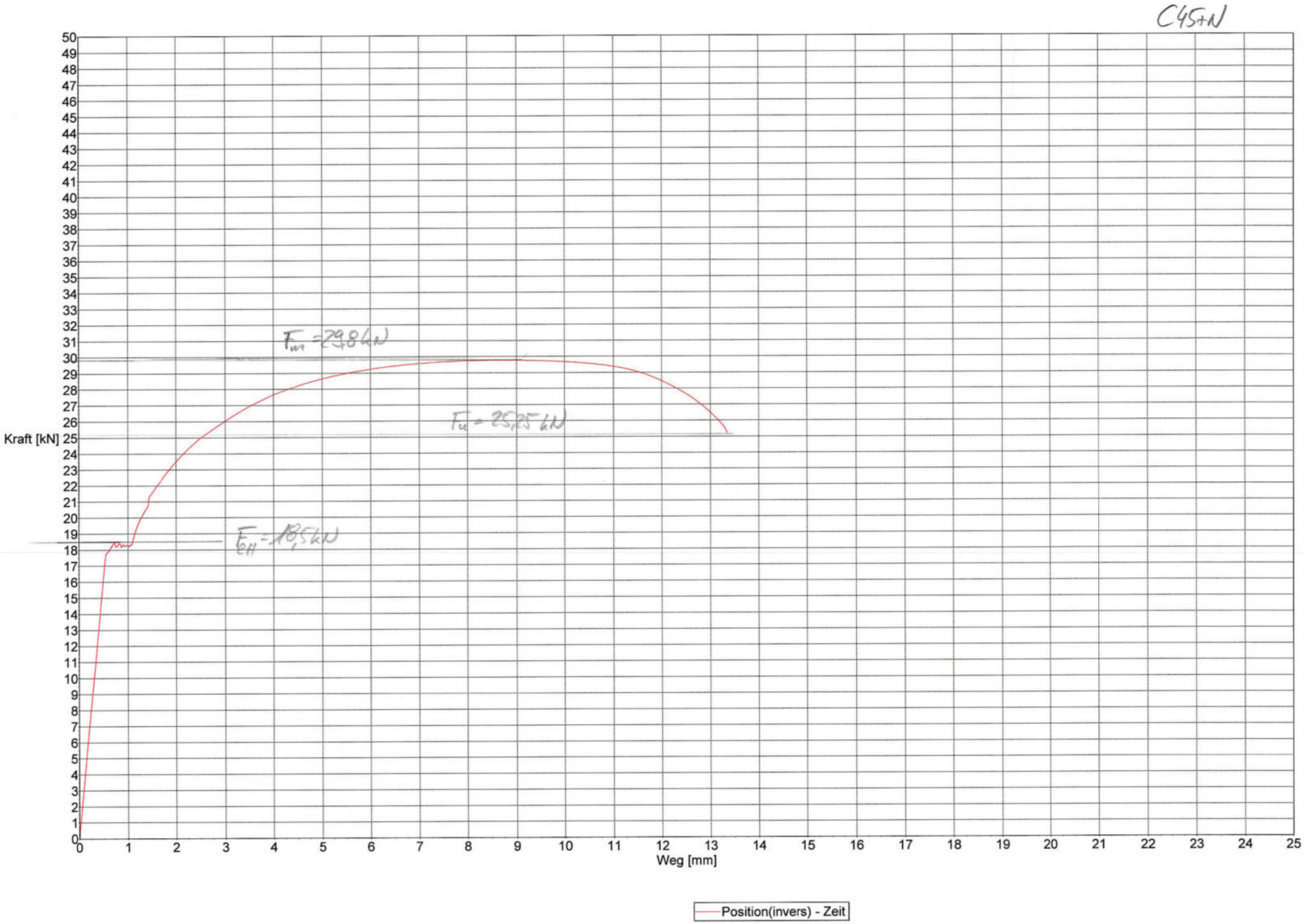

S235JR+C
Kraft [kN]
Weg [µm]
Kraft - Längenänderung (fein)
Rp0,2 = 24,25 kN / 50,2655 mm² = 482 MPa
80 µm
Fp0,2 = 24,25 kN
ΔF = 28 kN
E = (28000 N / 50,2655 mm²) / (80 µm / 30000 µm) = 209000 MPa
Le0,2
Le0,2 = 0,2 % · 30 mm = 6 mm% = 6/100 mm = 60 µm
Le = 30 mm = 30000 µm

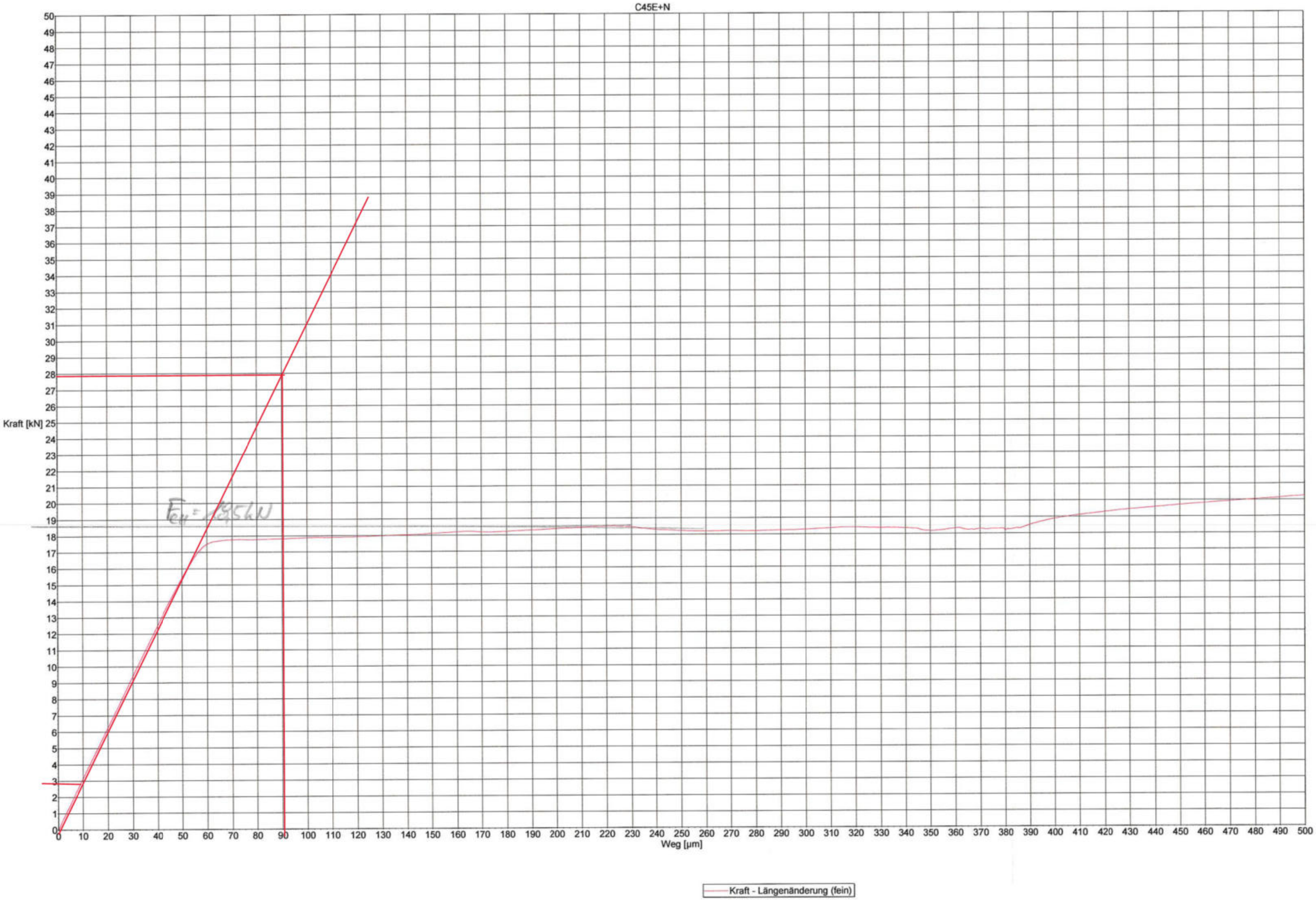
S235 JR+N



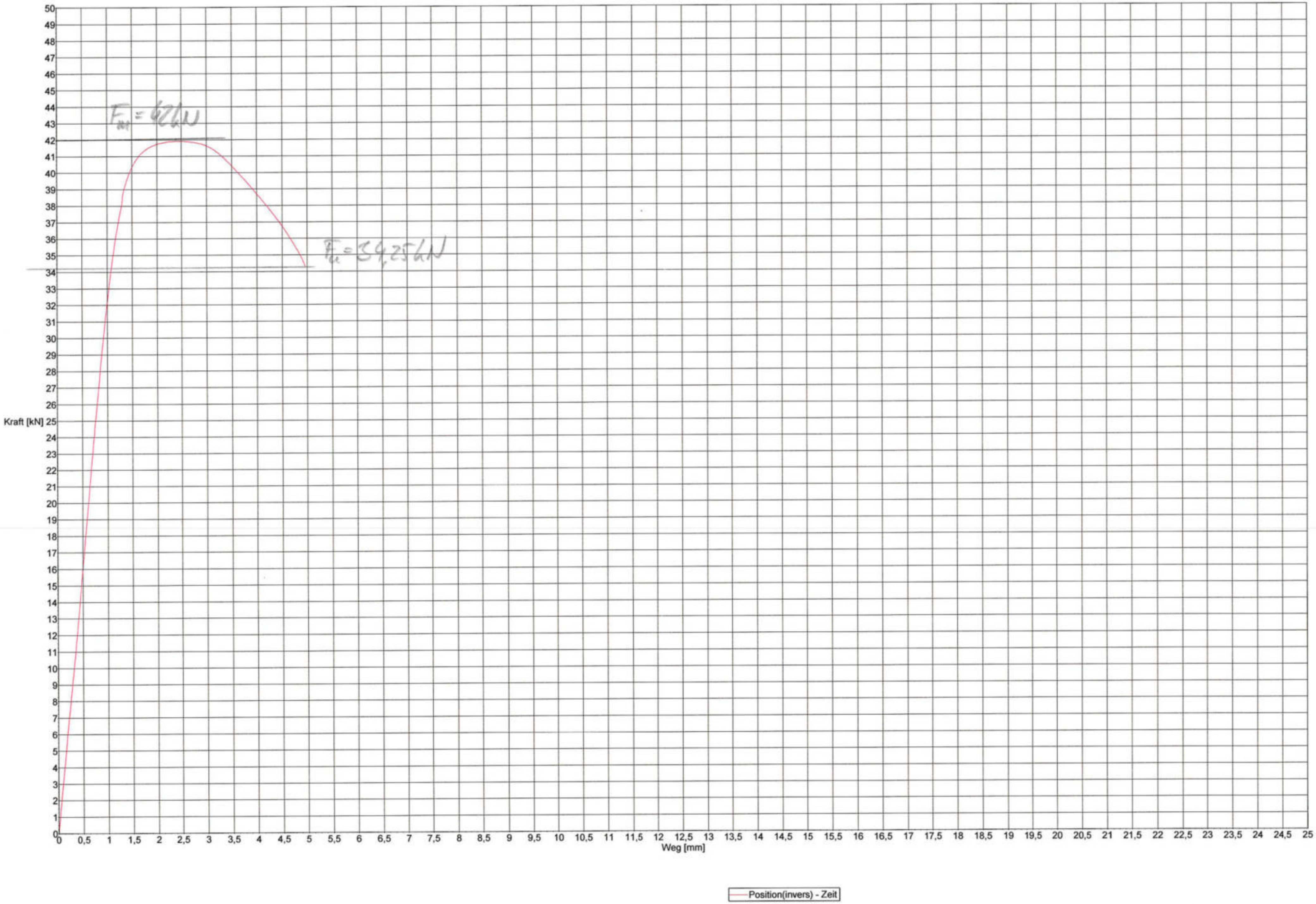


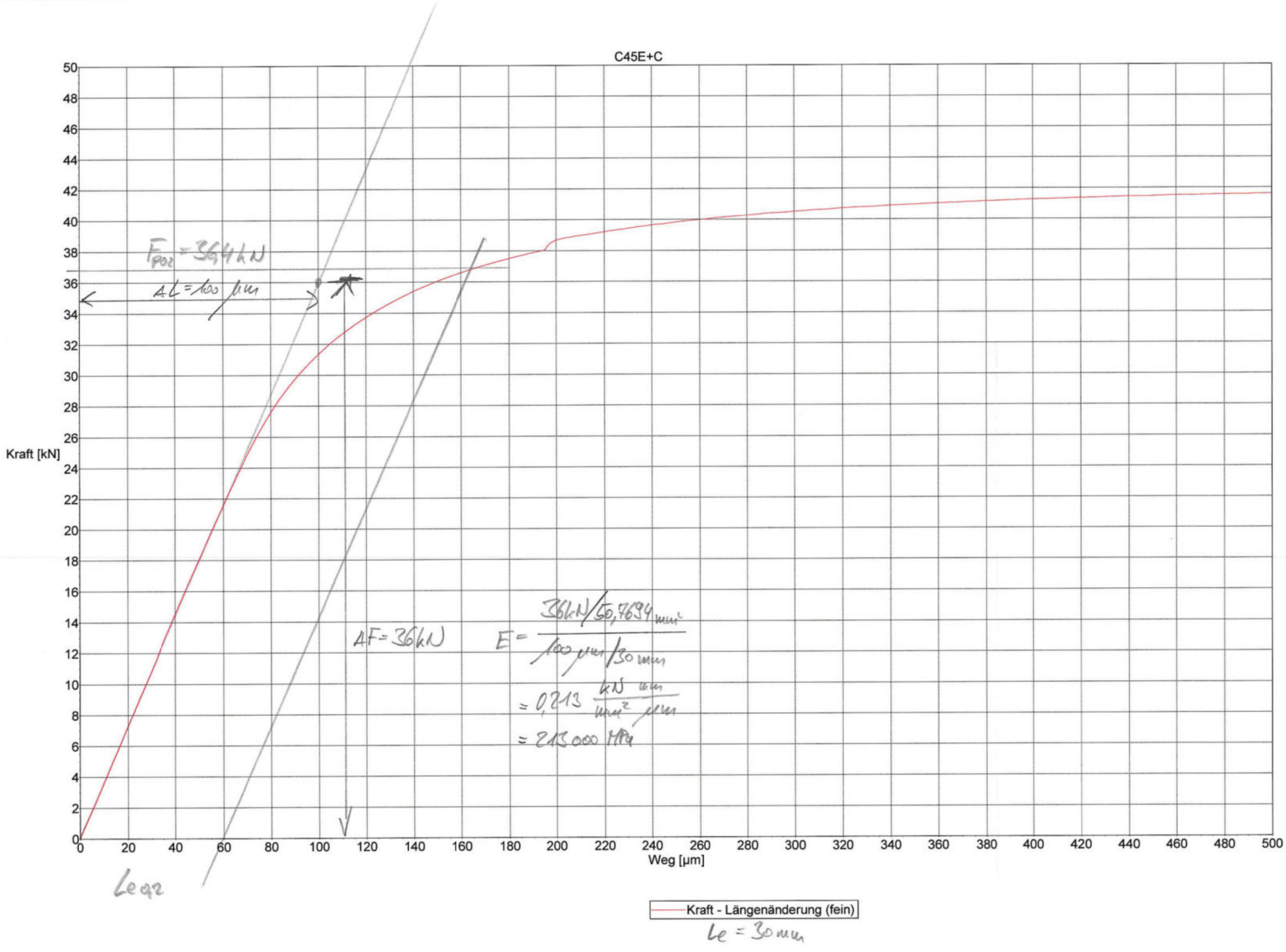
S235JR+N
Kraft [kN]
0
2
4
6
8
10
12
14
16
18
20
22
24
26
28
30
32
34
36
38
40
42
44
46
48
50
Weg [µm]
0
20
40
60
80
100
120
140
160
180
200
220
240
260
280
300
320
340
360
380
400
420
440
460
480
500
Kraft - Längenänderung (fein)
F_eH = ~~14,7~~ 15,50 kN
14 kN
E = (14000 N / 50,5171 mm²) / (40 µm / 30000 µm) = 208000 MPa
40 µm


C45+N
$F_m = 29{,}8\,kN$
$F_u = 25{,}25\,kN$
$F_{eH} = 18{,}5\,kN$
Kraft [kN]
Weg [mm]
Position(invers) - Zeit


C45E+N
Kraft [kN]
Weg [µm]
F_{eH} = 18,5 kN
Kraft - Längenänderung (fein)

C45+C




C45E+C
Kraft [kN]
Weg [µm]
Kraft - Längenänderung (fein)
$F_{p02} = 36,4\,kN$
$\Delta L = 100\,\mu m$
$\Delta F = 36\,kN$
$E = \frac{36\,kN / 50,7694\,mm^2}{100\,\mu m / 30\,mm}$
$= 0,213\,\frac{kN}{mm^2}\,\frac{mm}{\mu m}$
$= 213\,000\,MPa$
$L_{e\,q2}$
$L_e = 30\,mm$